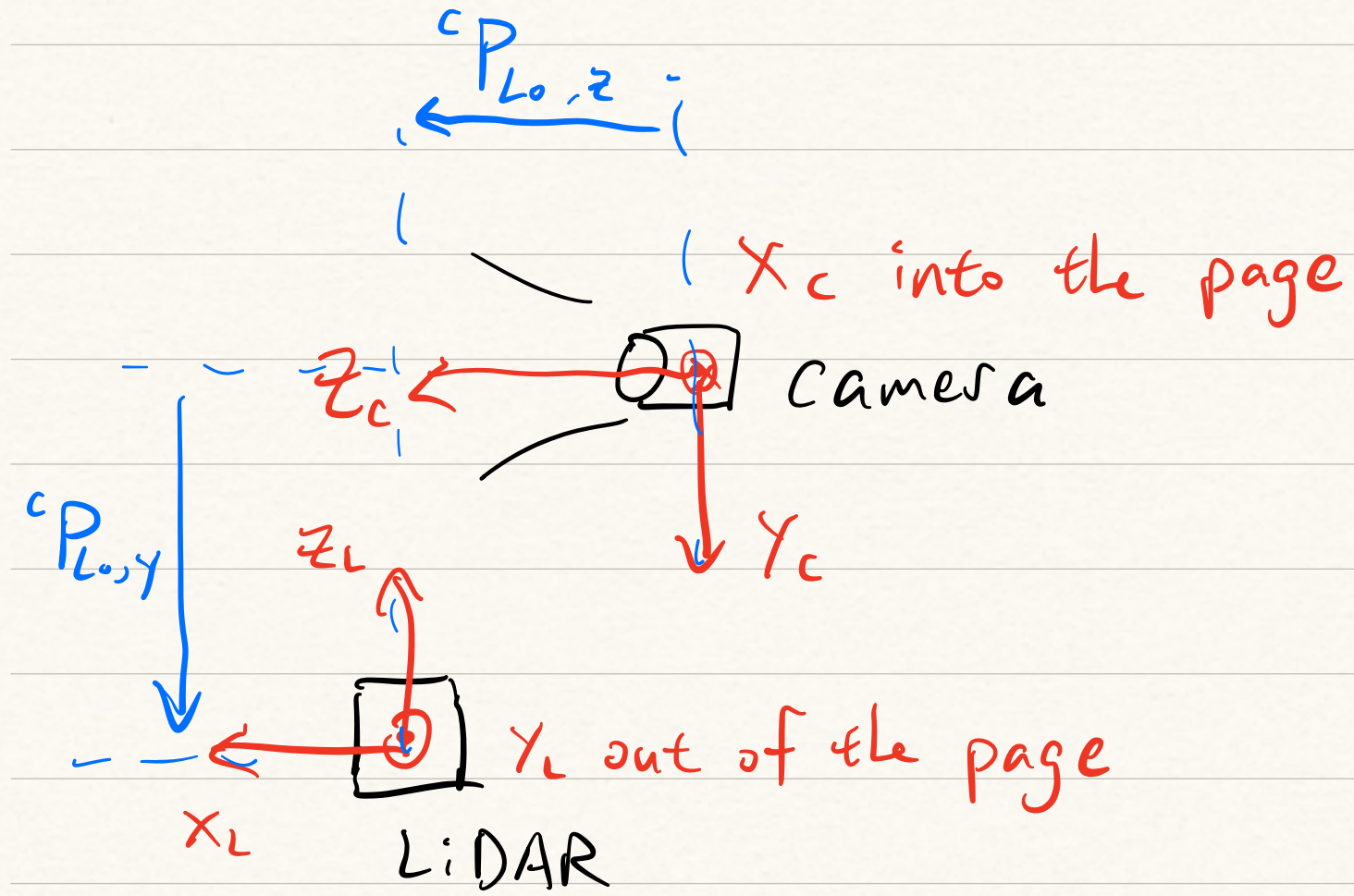


${}^{C}_{L}R$: rotation matrix that maps LiDAR frame points to camera frame's orientation.

$$\begin{aligned} X_c &= -Y_L \\ Y_c &= -Z_L \\ Z_c &= X_L \end{aligned} \rightarrow \begin{bmatrix} X_c \\ Y_c \\ Z_c \end{bmatrix} = \underbrace{\begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & 0 \end{bmatrix}}_{{}^{C}_{L}R} \begin{bmatrix} X_L \\ Y_L \\ Z_L \end{bmatrix}$$

${}^{C}P_{Lo}$: position of the origin of the LiDAR frame seen from the camera frame.

$$\overset{3\times1}{{}^{C}P_{Lo}} = \begin{bmatrix} {}^{C}P_{Lo,x} \\ {}^{C}P_{Lo,y} \\ {}^{C}P_{Lo,z} \end{bmatrix} \qquad \overset{3\times3}{{}^{C}_{L}T} = \begin{bmatrix} {}^{C}_{L}R & {}^{C}P_{Lo} \\ 0 & 1 \end{bmatrix}$$

0: 1×3 $\quad$ 1: 1×1

$${}^{C}P = {}^{C}_{L}T\ {}^{L}P$$

${}^{L}P$: points seen in LiDAR's frame

${}^{C}P$: points seen in Camera's frame.

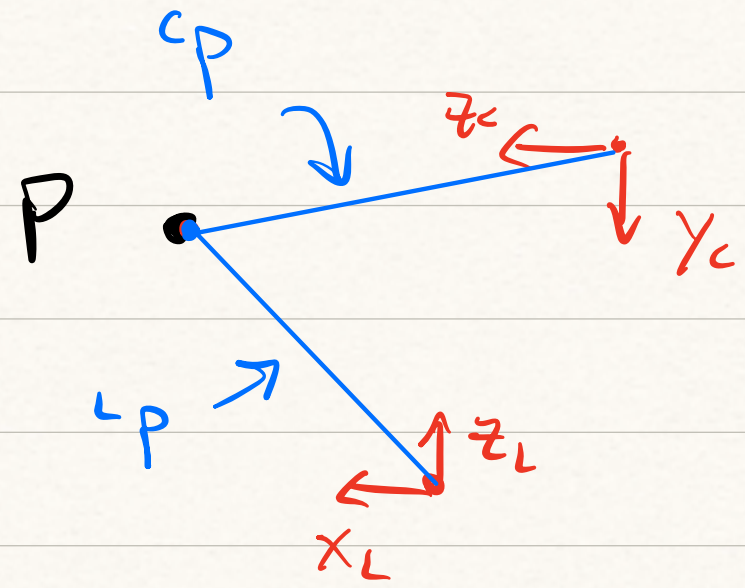
${}^{C}_{L}T$ completely maps points in LiDAR's frame into camera's frame.

$$\begin{bmatrix} {}^cP_x \\ {}^cP_y \\ {}^cP_z \\ 1 \end{bmatrix} = {}^c_L T \begin{bmatrix} {}^LP_x \\ {}^LP_y \\ {}^LP_z \\ 1 \end{bmatrix}$$

4×4 ↘

↙ has to have dimension of 4.

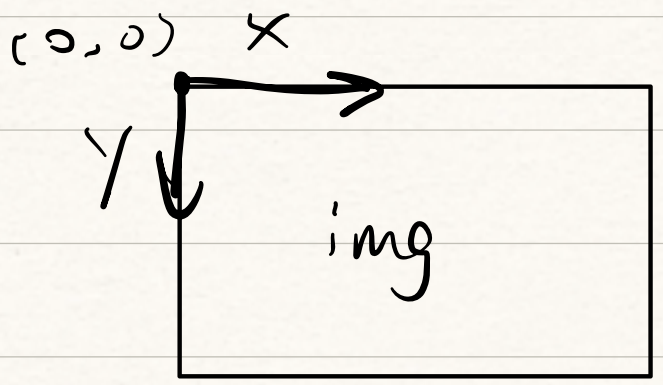
need to add 1 at the end, called homogeneous coordinates.



- Why camera frame look like this ?



z determined to be the depth.



once z is determined, if want x to point right, y has to point down (right-hand rule)

→ This is also the coordinate system used by OpenCV.

→ Be careful img[i, j] is still i^{th} row and j^{th} column. So img[y, x] gives the intensity value at (X, Y) pixel.